Palm OS Programming Recipes

Using NetLib Socket Calls

This recipe illustrates many of the basic socket calls supported by the Palm OS® Net library. It assumes you already have some basic knowledge of socket programming. Following these steps, you can create a server application that listens on a port and creates a data socket for communication with a client. For details on the Net library, see the "Network Communication" chapter in the *Palm OS Programmer's Companion*.

First you need to include the appropriate header files in your project:

```
#include <NetLib.h>
#include <sys_socket.h>
```

You also need to statically link `NetSocket.lib` in your project.

Before using the Net library functions, you need to load the Net library, as shown in the following code:

```
err = SysLibFind ("Net Lib", AppNetRefnum);
if (err != 0)
{
    /* If the library is not loaded, load the library*/
    err = SysLibLoad (sysFileTLibrary, sysFileCNet, &AppNetRefnum);
    if (err)
      /* display some error for the user */
```

Open the socket using `NetLibSocketOpen`.

```
MasterSocket = NetLibSocketOpen (AppNetRefnum, AF_INET, SOCK_STREAM, 0,
    AppNetTimeout, &errno);
```

Bind the socket to a local address using `NetLibSocketBind`.

```
AddrLength = sizeof(Addr);
bzero ((char *) &Addr, AddrLength);
Addr.sin_family = AF_INET;
Addr.sin_port = htons (Port);
NetLibSocketBind (AppNetRefnum, MasterSocket, (NetSocketAddrType *)&Addr,
    sizeof(struct sockaddr_in), AppNetTimeout, &errno);
```

Set the maximum pending connection-request queue length using `NetLibSocketListen`. Note that Palm OS currently only supports one client connection per device; any queue length value other than 1 will be ignored.

```
NetLibSocketListen (AppNetRefnum, MasterSocket,1, AppNetTimeout, &errno);
```

Accept a connection from a stream-based socket using `NetLibSocketAccept`. `NetLibSocketAccept` blocks until a successful connection request is obtained from a remote client. After a successful connection is made, `NetLibSocketAccept` returns the

address of the remote host and the socket descriptor of a new socket. Use the new socket to send and receive data. You should close the listen socket since you already have one client connected at this point.

```
NewSocket = NetLibSocketAccept (AppNetRefnum ,MasterSocket,
    (NetSocketAddrType* )&AddrNew, (Int16 *)&AddrLength,
    AppNetTimeout, &errno);
```

Note: `NetLibSocketAccept` blocks the user interface until the client requests to connect, which degrades the user experience. You should call `NetLibSocketAccept` with a small value for `timeout`. Place this call within a loop that checks for and processes events to ensure that your program complies with Palm's user interface guidelines.

Send and receive packets of data using `NetLibSend` and `NetLibReceive`.

```
RetVal = NetLibSend (AppNetRefnum, NewSocket,&buffer, bufferLen,0,0,0,
    AppNetTimeout , &errno);

RetVal = NetLibReceive (AppNetRefnum, NewSocket,&buffer,bufferLen,0,0,
    0, AppNetTimeout, &errno);
```

Close the socket using `NetLibSocketClose`. Note that `NetLibSocketClose` doesn't close the socket immediately. The system actually closes the socket sometime after the call to `NetLibSocketClose`. You can change this by setting the `netSocketOptSockLinger` option using the `NetLibSocketOptionSet` function.

```
NetLibSocketClose(AppNetRefnum, NewSocket, AppNetTimeout, &errno);
```